

Q:- A square matrix A is invertible iff $\lambda = 0$ is not an eigen value of A .

Solution:- Let A be a square matrix of order n .

Suppose that A is invertible i.e., A^{-1} exists

$$\Rightarrow \det A \neq 0$$

$$\Rightarrow \det (A - (0)I) \neq 0$$

$$\Rightarrow \det (A - \lambda I) \neq 0$$

i.e., $\lambda = 0$ is not a root of characteristic equation, hence $\lambda = 0$ is not an eigen value of A .

Conversely, Suppose that $\lambda=0$ is not an eigenvalue of matrix A .

$\Rightarrow \lambda=0$ is not a root of the characteristic equation

i.e, $\det(\lambda I - A) \neq 0$

$$\Rightarrow \det(0I - A) \neq 0$$

$$\Rightarrow \det(-A) \neq 0$$

$$\Rightarrow (-1)^n \det(A) \neq 0$$

$$\Rightarrow \det A \neq 0$$

Thus A is an invertible matrix.

Hence proved.